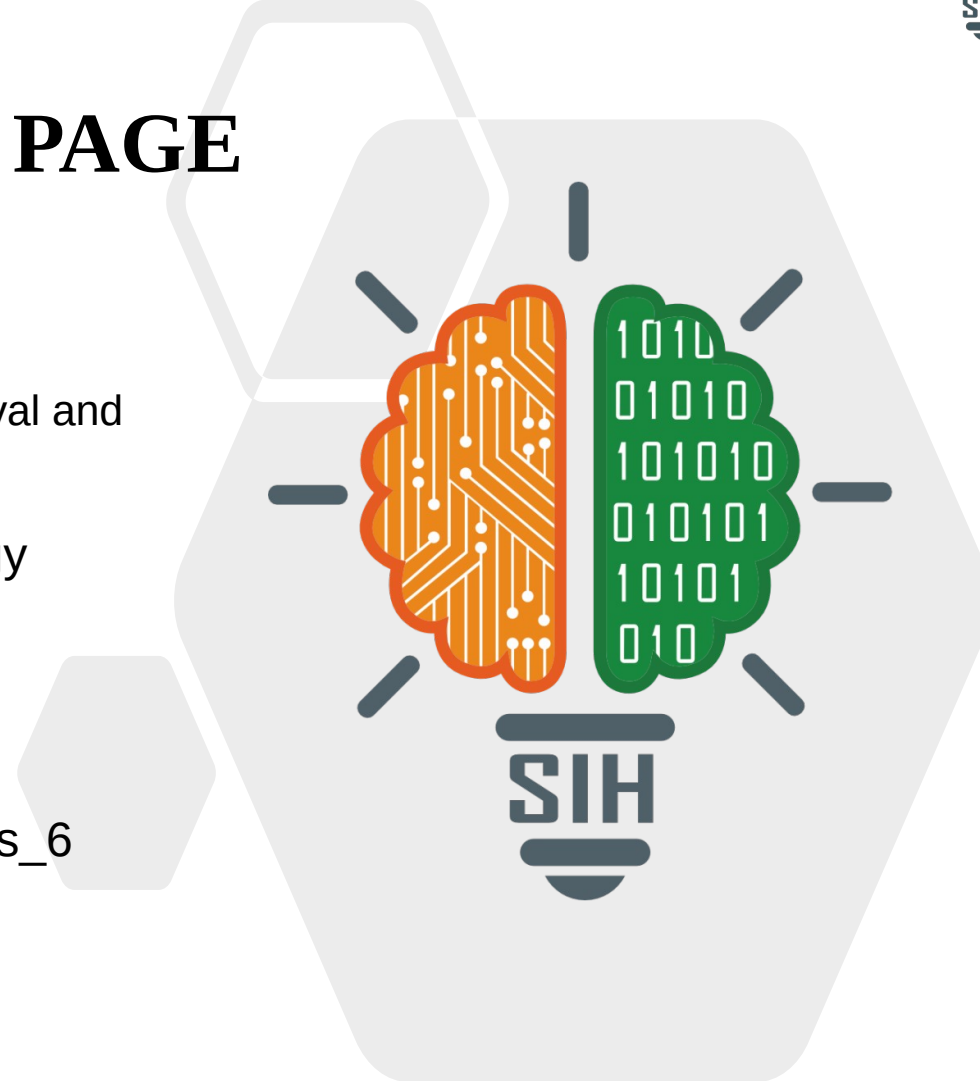




TITLE PAGE

- **Problem Statement ID** – 26240
- **Problem Statement Title** – AI-Based Spring Revival and Recharge Planning for Tribal Areas
- **Theme** – Agriculture, Food and Rural Technology
- **PS Category** – Software
- **Team ID** – 178601
- **Team Name (Registered on portal)** – TechDevs_6



JalSrot (Spring Revival Planning)

Problem:

- Where a spring's water comes from cannot be read off the surface; terrain alone misleads
- Finding recharge sites needs a hydrogeologist on foot, so remote tribal hamlets wait years

Our Solution:

- A web GIS tool that combines terrain, geology, rainfall, land use and spring records
- It marks the probable springshed and ranks sites for recharge work

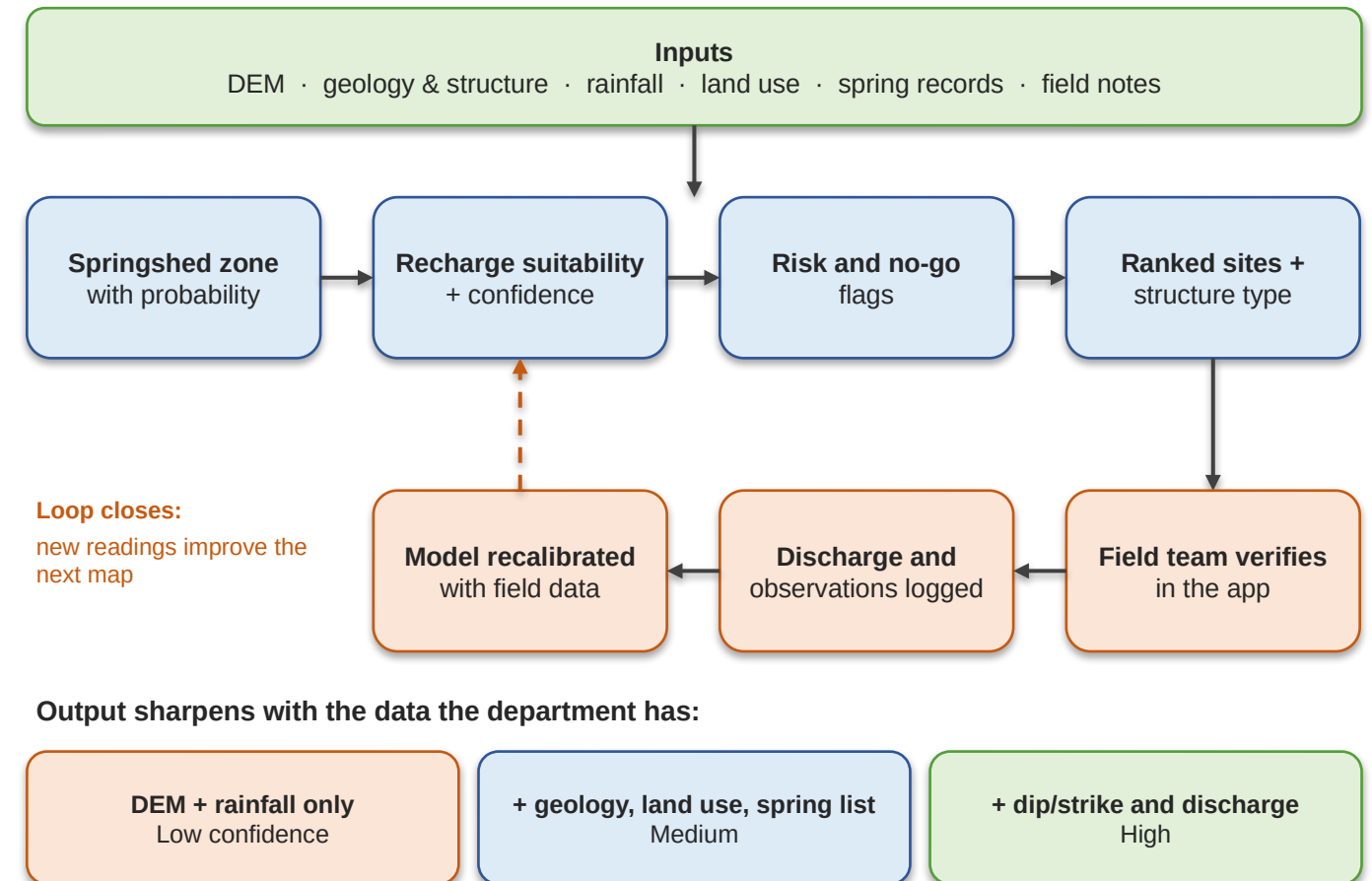
How it solves the problem:

- Experts verify a short ranked list, not every hillside
- Each map shows how sure we are, so thin data is never hidden
- Field readings go back into the model

Innovation & Uniqueness:

- Springshed follows rock dip and fractures, not just the watershed line
- Starts from DEM + rainfall alone and sharpens as data arrives
- Offline phone app records strike, dip and discharge in the village

System Flow:



TECHNICAL APPROACH

Technology Used:

Geo processing: Python, GDAL, rasterio, WhiteboxTools

ML: scikit-learn, XGBoost, SHAP

Backend: FastAPI, Celery + Redis

Database: PostgreSQL + PostGIS

Map serving: TiTiler (COG tiles), MinIO

Web GIS: Next.js, MapLibre GL

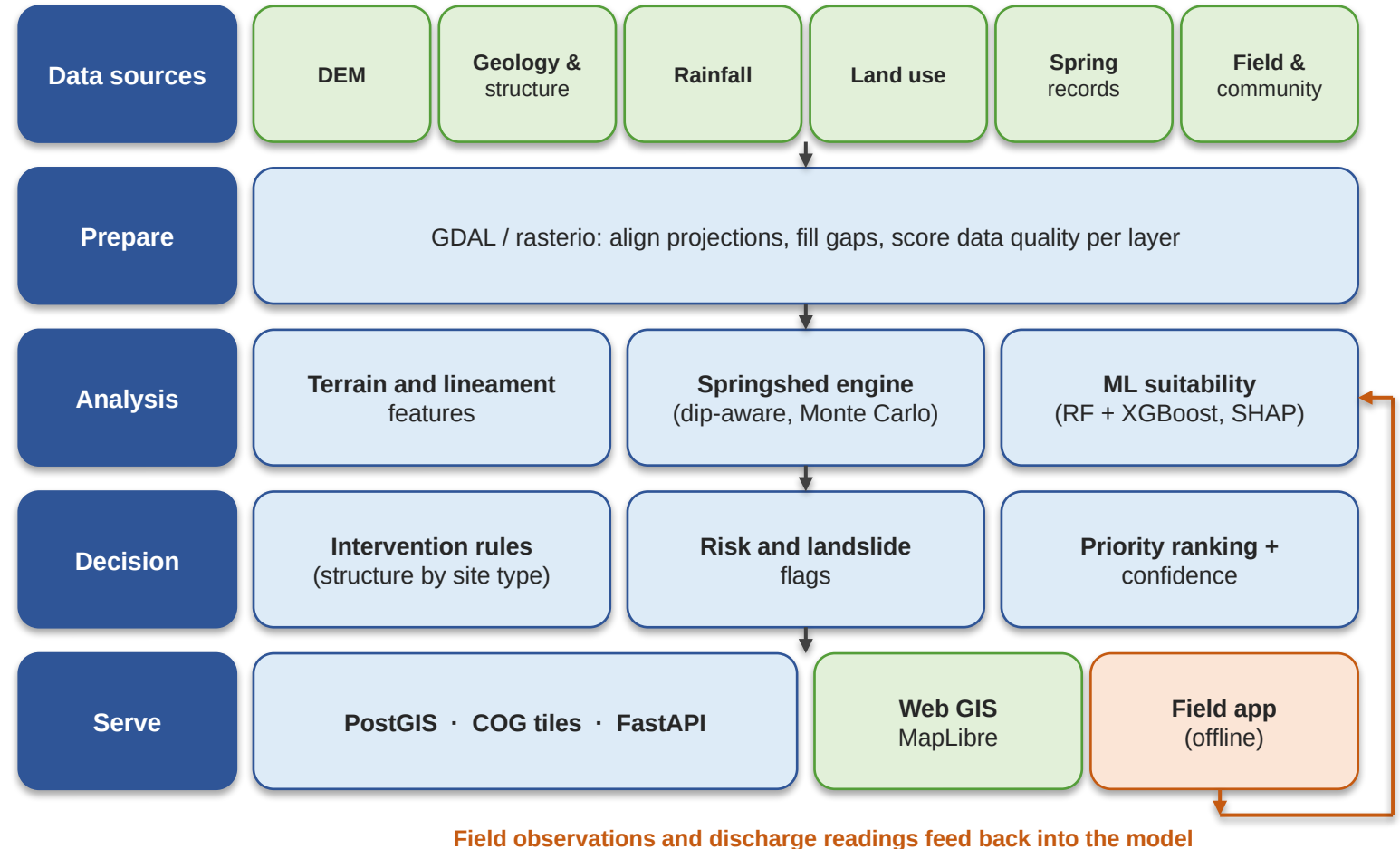
Field app: Capacitor Android / PWA, IndexedDB sync

Open data: SRTM / Cartosat DEM, GSI Bhukosh, IMD, Bhuvan land use

MLOps: MLflow, GitHub Actions

Language: Hindi and regional, via Bhashini

Architecture:



FEASIBILITY AND VIABILITY

Feasibility:

Technical: all inputs are open data and the stack is open source; one pilot springshed runs on a single server

Operational: field team needs only an Android phone; works offline in the village

Economic: no licence cost; the phone compass replaces survey equipment for strike and dip

Data: Bhuvan, GSI, IMD and India-WRIS; outputs as GeoTIFF / GeoJSON and OGC map services

Rollout: one block (20-30 springs) → district → state

Pilot plan: verify top-ranked sites in the monsoon, log discharge through the next dry season, compare against the model

Challenges & Risks:	Strategies to Overcome:
Few labelled springs	Expert-weighted fallback; map marked low confidence
Coarse DEM in some areas	Resolution stated on the map; confidence drops
No structural measurements	Lineaments inferred from the DEM; field team adds readings
Map trusted too much	Every output says "indicative, verify on site"
Landslide-prone slopes	Separate risk layer; no-go flag on the site
Low field adoption	Four-step app, voice notes, local language

IMPACT AND BENEFITS

Today vs with JalSrot:

	Today	With JalSrot
Find recharge area	Field survey, spring by spring	Probable zone from data, then verified
Choose sites	Expert judgement, case by case	Ranked list with reasons
Know the risk	Found after the structure is built	Flagged before work starts
Learn from results	Discharge rarely recorded	Logged and fed back to the model
Reach remote areas	Waits for an expert visit	Pre-screened; team visits the best sites first

Benefits:

Field officer: knows which sites to visit first

Hydrogeologist: time goes into verifying, not scanning

District / department: a ranked list with reasons to support funding

Tribal community: more dependable water in the dry months

Impacts:

Social: less time spent fetching water, mostly by women and children

Economic: fewer structures built where they do not work

Environmental: right aquifers recharged; no works on unstable slopes

System: a growing spring-discharge record for the department

RESEARCH AND REFERENCES

S. No	Title	Published by	Year	Used for
1	SRTM Digital Elevation Model	NASA / USGS	2000	Slope, aspect, drainage, flow paths
2	Bhuvan land use / land cover and Cartosat DEM	ISRO – NRSC	2022	Land cover and finer terrain
3	Bhukosh geological maps	Geological Survey of India	2021	Rock types, contacts, faults
4	CHIRPS rainfall dataset (Funk et al.)	UC Santa Barbara CHG	2015	Rainfall trends
5	India-WRIS	Ministry of Jal Shakti	2010	Drainage, groundwater context
6	Inventory and Revival of Springs in the Himalayas	NITI Aayog	2018	Springshed approach, revival practice
7	Random Forests (Breiman)	Machine Learning journal	2001	Recharge suitability model
8	SHAP – A Unified Approach to Interpreting Model Predictions	Lundberg & Lee, NeurIPS	2017	Explaining each site score
9	WhiteboxTools (Lindsay)	University of Guelph	2016	Terrain and hydrological analysis